Table 1: Changes in the 11-task language average at each training step, with 95% intervals. Changes are a checkpoint minus the checkpoint it was trained from, except those marked ⇒, which span several training steps. Scores are percent accuracy with the paper's metrics (normalized accuracy for ARC-C, ARC-E, HellaSwag, OBQA and PIQA, accuracy otherwise), under the prompt-example seed the paper's tables use. Intervals are 95% paired bootstrap intervals over the questions, resampled within each task, with 2,000 replicates. q is the Benjamini-Hochberg corrected p-value over the average changes of each model. The last column repeats the change with each question averaged over the three prompt-example seeds. The seeds change only the few-shot examples, so the intervals describe the sample of questions, not training variation. FineWeb-Edu's 16K extended checkpoint is the one its 16K SFT checkpoint was trained from. Nemotron-CC-HQ's checkpoints are those of Table **??**. No change starts from its 16K extended checkpoint, which is not the model its SFT checkpoint was trained from.

Model	Change	From	To	Change [95% CI]	q	Three seeds
MixtureVitae	base, 4K → base, 16K	55.82	54.96	−0.87 [−1.56, −0.15]	0.042	−0.96 [−1.61, −0.24]
	base, 16K → SFT	54.96	55.45	+0.49 [−0.12, +1.13]	0.167	+0.50 [−0.08, +1.10]
	SFT → SFT, DPO	55.45	55.51	+0.06 [−0.01, +0.13]	0.152	+0.10 [+0.04, +0.16]
	base, 16K → OT30K	54.96	55.64	+0.68 [+0.30, +1.10]	0.005	+0.68 [+0.33, +1.03]
	SFT → SFT, OT30K	55.45	55.68	+0.23 [−0.23, +0.66]	0.344	+0.34 [−0.09, +0.77]
	SFT, DPO → SFT, DPO, OT30K	55.51	55.74	+0.23 [−0.24, +0.67]	0.344	+0.27 [−0.17, +0.69]
	base, 16K ⇒ SFT, DPO	54.96	55.51	+0.56 [−0.07, +1.18]	0.152	+0.60 [+0.01, +1.20]
	base, 16K ⇒ SFT, OT30K	54.96	55.68	+0.72 [+0.12, +1.35]	0.042	+0.84 [+0.26, +1.42]
	base, 16K ⇒ SFT, DPO, OT30K	54.96	55.74	+0.78 [+0.19, +1.41]	0.042	+0.86 [+0.28, +1.44]
Comma0.1	base, 4K → base, 16K	50.05	52.10	+2.05 [+1.38, +2.76]	0.002	+2.00 [+1.37, +2.67]
	base, 16K → SFT	52.10	51.15	−0.95 [−1.58, −0.29]	0.005	−0.82 [−1.41, −0.21]
	SFT → SFT, DPO	51.15	51.34	+0.20 [+0.04, +0.42]	0.007	+0.21 [+0.08, +0.42]
	SFT → SFT, OT30K	51.15	51.30	+0.16 [−0.42, +0.71]	0.612	+0.21 [−0.32, +0.70]
	SFT, DPO → SFT, DPO, OT30K	51.34	50.21	−1.14 [−1.79, −0.47]	0.004	−1.30 [−1.89, −0.69]
	base, 16K ⇒ SFT, DPO	52.10	51.34	−0.75 [−1.37, −0.13]	0.018	−0.61 [−1.18, −0.04]
	base, 16K ⇒ SFT, OT30K	52.10	51.30	−0.80 [−1.45, −0.18]	0.010	−0.61 [−1.26, −0.06]
	base, 16K ⇒ SFT, DPO, OT30K	52.10	50.21	−1.89 [−2.55, −1.28]	0.002	−1.91 [−2.52, −1.33]
Nemotron-CC-HQ	base, 4K → base, 16K	64.17	64.35	+0.18 [−0.29, +0.65]	0.459	+0.12 [−0.32, +0.57]
	SFT → SFT, DPO	62.43	62.53	+0.10 [+0.04, +0.15]	0.002	+0.12 [+0.08, +0.16]
	SFT → SFT, OT30K	62.43	61.90	−0.53 [−1.00, −0.10]	0.045	−0.38 [−0.80, +0.02]
	SFT, DPO → SFT, DPO, OT30K	62.53	62.22	−0.31 [−0.78, +0.10]	0.207	−0.31 [−0.72, +0.09]
FineWeb-Edu	base, 4K → SFT, 4K	56.75	57.37	+0.62 [+0.07, +1.15]	0.081	+0.79 [+0.26, +1.27]
	SFT, 4K → SFT, DPO, 4K	57.37	57.46	+0.09 [+0.03, +0.15]	0.012	+0.08 [+0.04, +0.14]
	base, 4K → base, 16K	56.75	57.30	+0.55 [+0.00, +1.10]	0.097	+0.63 [+0.12, +1.17]
	base, 16K → SFT	57.30	56.78	−0.52 [−1.11, +0.04]	0.105	−0.47 [−1.02, +0.07]
	SFT → SFT, DPO	56.78	56.84	+0.06 [+0.01, +0.11]	0.070	+0.06 [+0.02, +0.10]
	SFT → SFT, OT30K	56.78	57.12	+0.34 [−0.13, +0.81]	0.164	+0.31 [−0.14, +0.77]
	base, 16K ⇒ SFT, DPO	57.30	56.84	−0.46 [−1.05, +0.09]	0.139	−0.41 [−0.96, +0.13]
	base, 16K ⇒ SFT, OT30K	57.30	57.12	−0.18 [−0.80, +0.46]	0.567	−0.15 [−0.77, +0.45]

Table 2: MixtureVitae: change on each language task at each training step, in points, and the 11-task average with two decimals. Each task cell is the estimated change, and Table 1 gives the interval of the average. Bold marks a change with $q < 0.05$ after the Benjamini-Hochberg correction over the model's per-task changes. Same records and conventions as Table 1.

Change	ARC-C	ARC-E	BoolQ	CSQA	COPA	HellaS	LAMBADA	MMLU	OBQA	PIQA	WinoG	Avg
base, 4K → base, 16K	+1.4	+0.5	−0.9	−2.5	−4.0	**−1.4**	**+1.2**	−0.9	−2.4	−0.7	+0.2	−0.87
base, 16K → SFT	**−2.5**	−1.2	+0.3	+1.6	+2.0	**+1.8**	+0.6	+0.8	**+2.4**	+0.1	−0.5	+0.49
SFT → SFT, DPO	+0.0	**+0.3**	−0.1	−0.1	+0.0	**+0.1**	+0.2	+0.1	+0.2	−0.1	−0.1	+0.06
base, 16K → OT30K	+0.9	−0.5	**+1.9**	+0.3	+2.0	**+1.0**	−0.7	+0.2	+1.4	+0.4	+0.7	+0.68
SFT → SFT, OT30K	+0.2	+0.1	+0.6	−0.7	+2.0	+0.4	**−1.0**	−0.6	+0.2	+0.3	+1.2	+0.23
SFT, DPO → SFT, DPO, OT30K	+0.5	−0.5	+0.4	+0.1	+2.0	**+0.4**	**−1.3**	**−0.7**	−0.6	+0.6	+1.5	+0.23
base, 16K ⇒ SFT, DPO	**−2.5**	−0.9	+0.3	+1.5	+2.0	**+1.9**	+0.8	+0.9	**+2.6**	+0.1	−0.6	+0.56
base, 16K ⇒ SFT, OT30K	**−2.3**	−1.1	+0.9	+0.8	+4.0	**+2.1**	−0.4	+0.2	**+2.6**	+0.4	+0.7	+0.72
base, 16K ⇒ SFT, DPO, OT30K	−2.0	**−1.4**	+0.7	+1.6	+4.0	**+2.3**	−0.5	+0.3	+2.0	+0.7	+0.9	+0.78

Table 3: All contrasts of the front-loading study on the 11-task language average, as Table **??** gives them for reasoning. The checkpoints are the 4K pretrained bases and their 16K extended checkpoints, before Tulu3 alignment, the sixteen of Table **??**. Intervals are 95% paired bootstrap intervals over the questions, resampled within each task, with 2,000 replicates. q-values are Benjamini–Hochberg adjusted within each family. Scores use the prompt-example seed of the language tables. Averaging each question over the three prompt-example seeds of each checkpoint moves no estimate by more than 0.38 points and moves no contrast across $q = 0.05$. The seeds change only the few-shot examples, so the intervals describe the sample of questions, not training variation.

	4K			**16K**		
Contrast	**Estimate**	**95% CI**	q	**Estimate**	**95% CI**	q
Front-loading effect, within source pool						
MixtureVitae	+4.78	[+3.94, +5.60]	< 0.001	+4.12	[+3.26, +4.99]	< 0.001
Comma0.1	+6.57	[+5.76, +7.39]	< 0.001	+4.93	[+4.09, +5.75]	< 0.001
Nemotron-CC-HQ	+0.38	[−0.42, +1.21]	0.377	+0.51	[−0.38, +1.34]	0.258
FineWeb-Edu	+6.10	[+5.31, +6.84]	< 0.001	+5.27	[+4.51, +6.01]	< 0.001
Permissive minus non-permissive source pool, both front-loaded						
MV − Nemotron	−8.73	[−9.65, −7.83]	< 0.001	−9.62	[−10.56, −8.65]	< 0.001
MV − FineWeb	−7.02	[−7.99, −6.05]	< 0.001	−7.62	[−8.53, −6.73]	< 0.001
Comma − Nemotron	−7.94	[−8.97, −6.93]	< 0.001	−7.55	[−8.50, −6.63]	< 0.001
Comma − FineWeb	−6.23	[−7.18, −5.26]	< 0.001	−5.54	[−6.42, −4.67]	< 0.001
Interaction: front-loading effect, permissive minus non-permissive						
MV − non-permissive	+1.53	[+0.47, +2.50]	0.010	+1.22	[+0.15, +2.33]	0.028
Comma − non-permissive	+3.33	[+2.33, +4.35]	0.001	+2.04	[+1.06, +3.09]	0.001